

ICMP PACKET FLOW IN A SWITCH

PING MESSAGE

WHY ARP GENERATED BEFORE ICMP?



ICMP + ARP



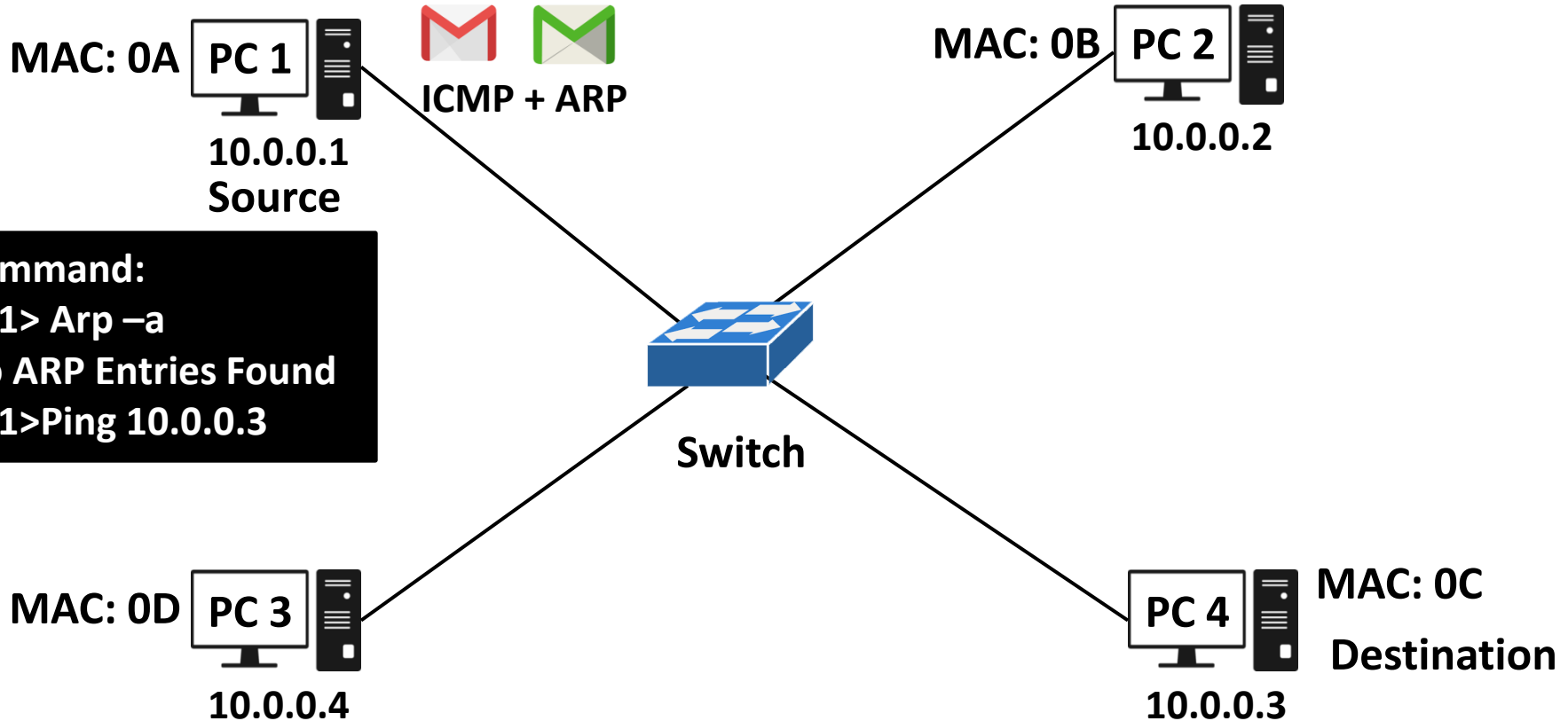
Switch:

- Works on Data Link Layer
- Send or Receive Data Frame
- MAC Address Ageing Time 300 Sec
- Maintain MAC Address Table
- Switch Does Broadcast 1st Time then Unicast
- Switch has CLI and GUI Mode
- Single Broadcast Domain
- Per Port Collision
- No. Of Ports: 16, 24, 32, 48
- Full Duplex Device

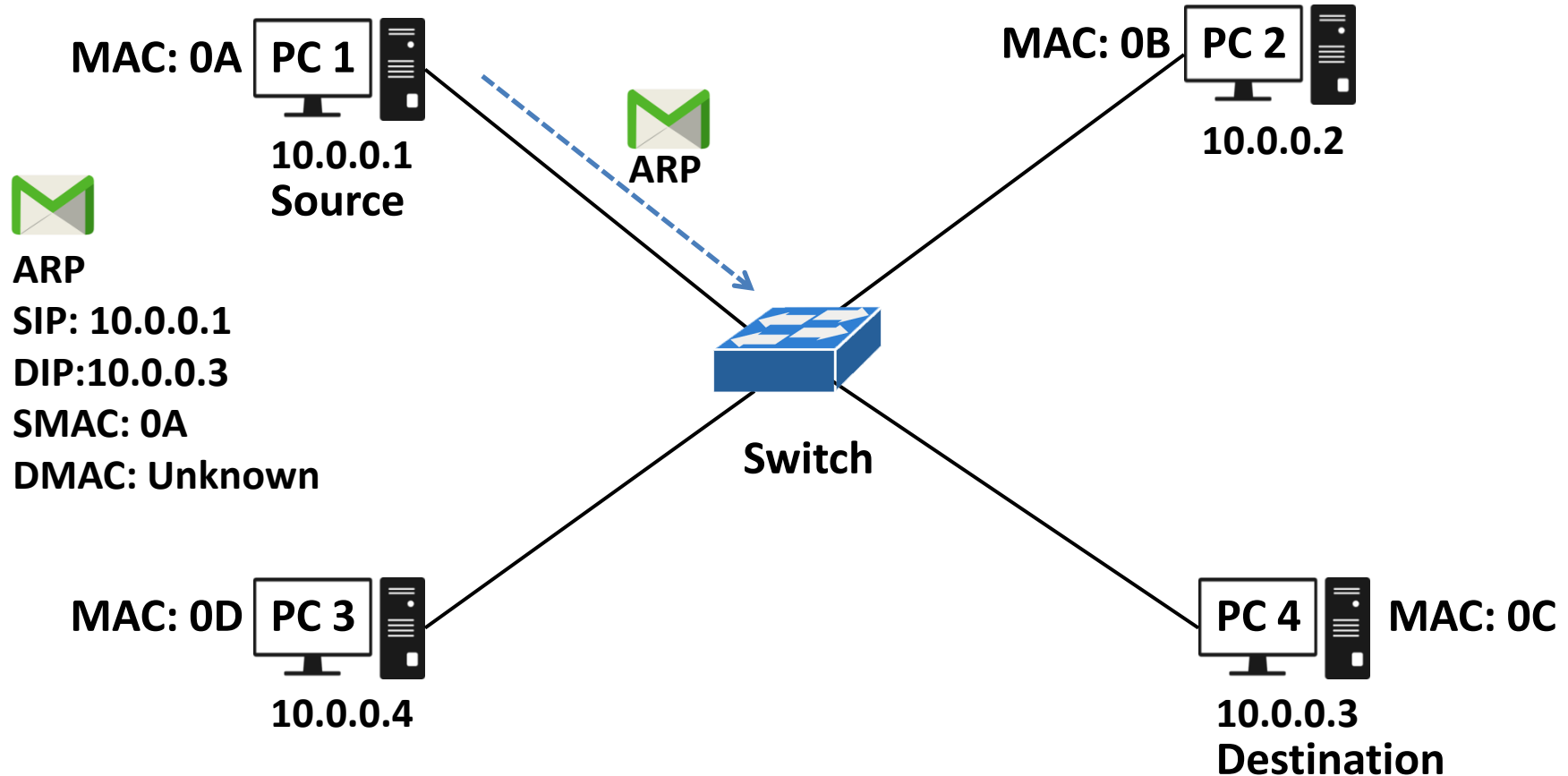


Switch

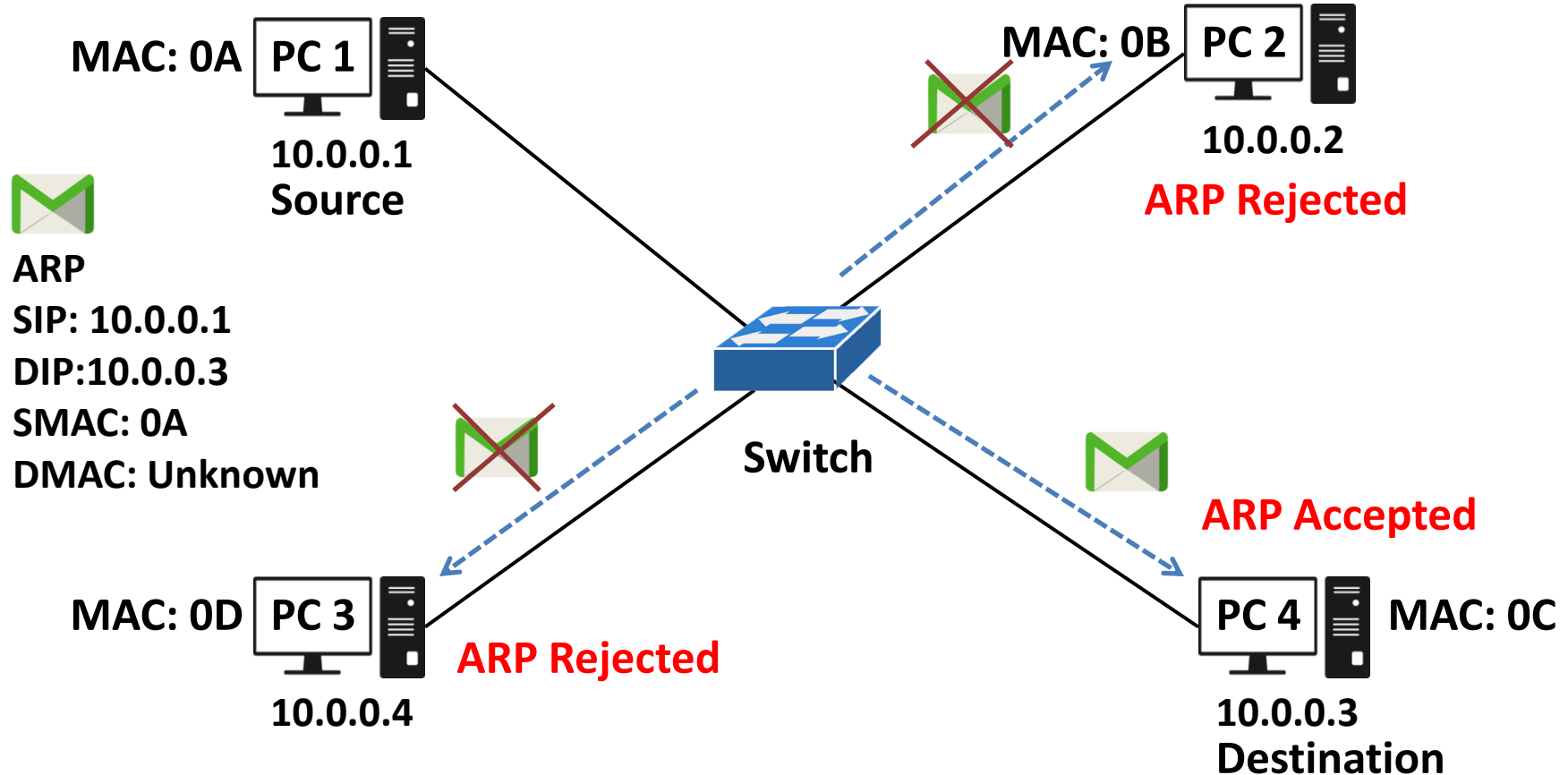
Source Doesn't have information of Destination MAC, So ARP Will be Generated Before ICMP to Get Destination MAC



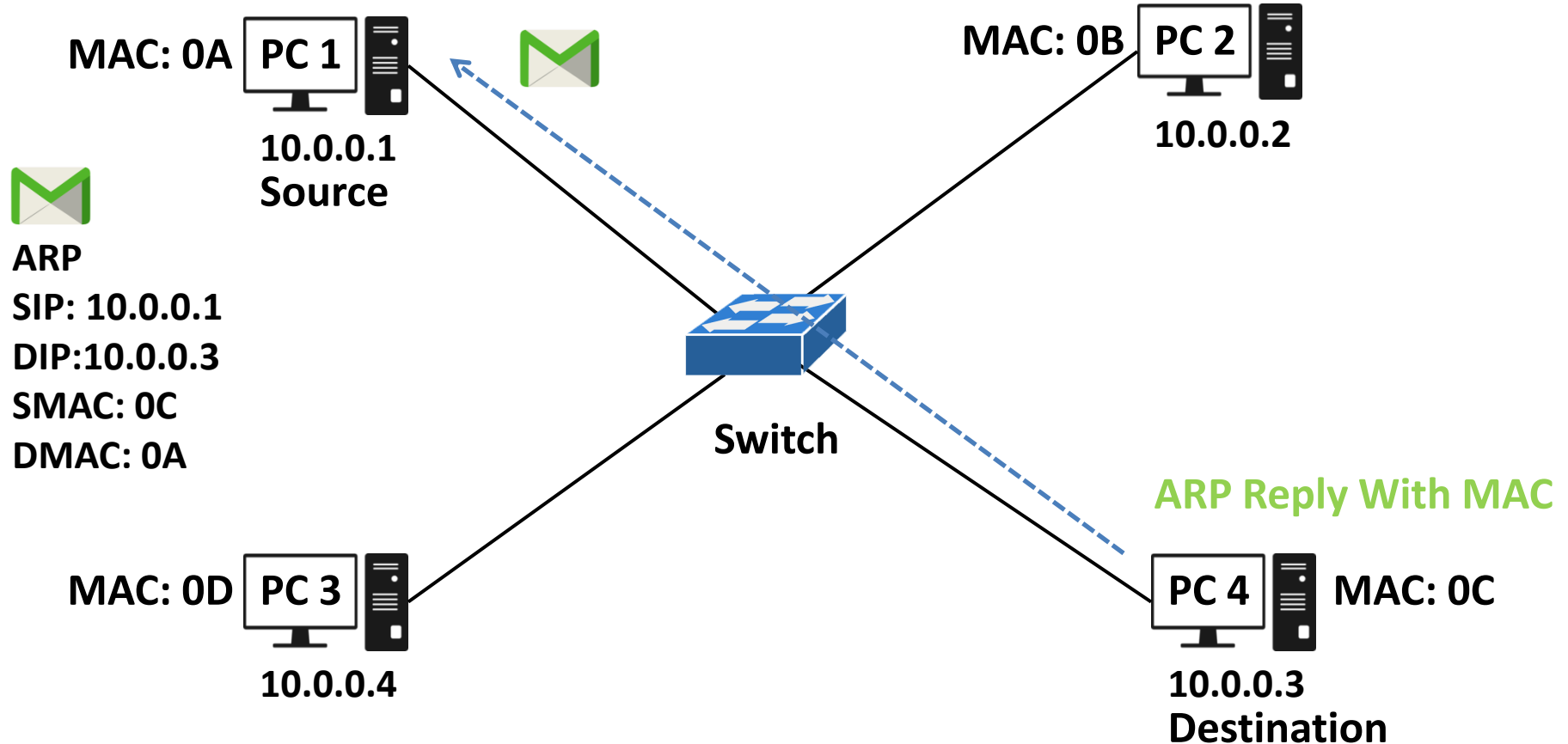
Firstly, ARP Protocol used to find MAC Address of Destination:



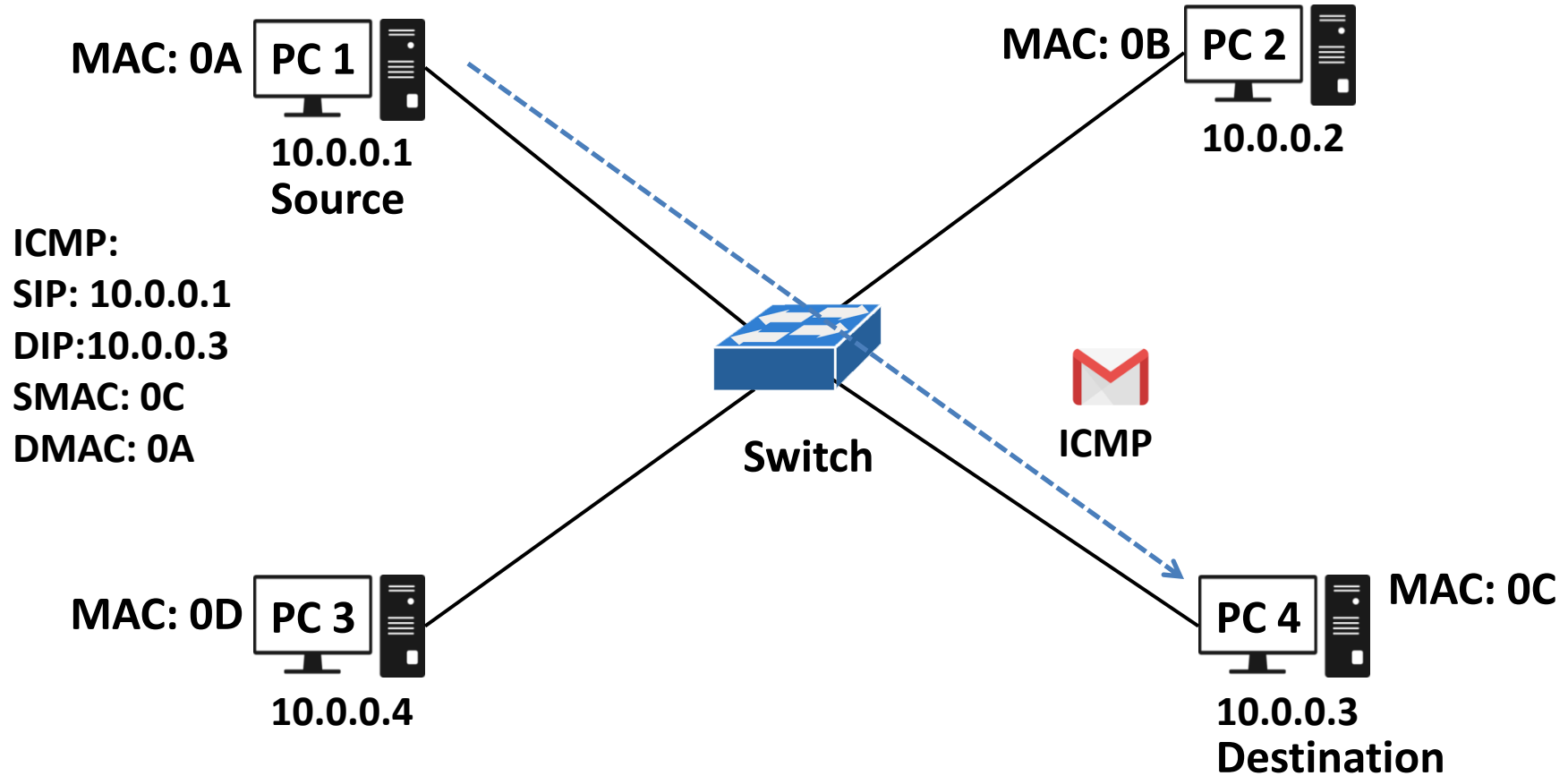
Switch Does Broadcast for First Time Only:



After Getting MAC Switch Does Unicast



ICMP 4 Packets will be forwarded one by one with size = 32 Bytes Each



In Response of ICMP packet the Acknowledgement will be received

